

UAV Trajectory Optimization Based on Pointer Networks and Adaptive Region Partitioning

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Abstract—uncrewed aerial vehicles (UAVs), characterized by their agility, affordability, and flexible deployment, exhibit significant advantages in scenarios such as disaster monitoring, target tracking, and environmental data collection. However, the limited onboard energy of UAVs poses a major challenge for long-duration or large-scale missions. To address this issue, this article proposes a dynamic trajectory planning framework for cooperative task search involving multiple UAVs. First, an UAV capability evaluation approach is developed to assess the relative performance of heterogeneous UAVs. Next, a density-aware clustering mechanism is employed to partition the search region based on spatial distance and regional density. After clustering, a sequential matching strategy is employed to assign UAVs with higher capabilities to larger or more complex task regions, ensuring efficient resource utilization. The problem is then formulated as a combinatorial optimization task, and a pointer network is designed to generate UAV trajectories. The network is trained using deep reinforcement learning (DRL) to produce near-optimal paths, thereby minimizing the overall system cost. The experimental results confirm that the proposed method can substantially lower total task execution expenditure.

Index Terms—Deep reinforcement learning (DRL), dynamic trajectory planning, pointer network, uncrewed aerial vehicle (UAV).

I. INTRODUCTION

WITH the swift advancement of Internet of Things (IoT) systems and digital technologies, the vision of an interconnected world—commonly referred to as the Internet of Everything—is gradually becoming a reality [1]. This trend is accompanied by an exponential increase in the number of network-connected devices and a growing demand for computational resources, posing significant challenges to existing network architectures [2]. As predicted by IDC, the worldwide count of mobile terminals is projected to hit 41.6 billion by 2025, producing nearly 79.4 ZB of digital information. Meanwhile, the number of interconnected devices accessing 5G networks is projected to reach 25 billion by the same year

[3]. Meanwhile, the surging proliferation of mobile terminals, along with the massive volume of data generated, places immense pressure on terrestrial network infrastructures, which are often limited in computing and storage capacity. This limitation becomes particularly critical for latency-sensitive applications, such as video conferencing, telemedicine, and autonomous driving, which require ultralow communication delay. In the future, a single base station may need to support an exponentially increasing number of computational nodes. As a result, mobile terminals will demand more competitive access networks with enhanced scalability, mobility, security, and intelligence [4].

At present and in the foreseeable future, mobile communication faces the following three major challenges.

- 1) The explosive growth of data, the diversity of content requests, and the increasing complexity of computational tasks. In the early stages of 5G architecture, base stations merely served as intermediaries for information exchange with the access network. However, the rapid increase in devices and traffic has greatly intensified pressure on cellular infrastructure, hindering base stations' ability to respond to user requests within the required time frame. This fails to fulfill the essential demands of intelligent 5G applications, including extensive coverage, strong reliability, and minimal latency [5].
- 2) The limitations of fixed infrastructure with constrained computing and storage resources. As user demand continues to grow and application scenarios become more dynamic, stationary base stations often lack the flexibility and capacity to support real-time, data-intensive services for mobile devices.
- 3) Limited energy supply, constrained communication resources, and frequent network congestion at the network edge [6]. In particular, under high-density access scenarios or unexpected events, traditional network infrastructures often struggle to respond promptly, leading to service interruptions and performance degradation, which critically undermines the quality of service and user experience for latency-sensitive applications.

Against this backdrop, uncrewed aerial vehicles (UAVs), owing to their strong maneuverability, affordability, and flexible deployment capabilities, are increasingly regarded as a promising complement to traditional infrastructure in addressing the aforementioned challenges [7]. In scenarios such as

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disaster monitoring, emergency communications, target search, and rescue operations, conventional communication facilities often fail to be deployed rapidly or are susceptible to damage. UAVs, by contrast, can swiftly reach target regions and perform real-time data acquisition and search tasks, making them particularly suitable for environments characterized by wide spatial distribution, complex terrain, or limited accessibility [8]. Beyond communication tasks, the integration of onboard computing units into UAVs has enabled their deployment in dynamic scenarios with rapidly increasing computational demands, such as disaster response, large-scale temporary outdoor events, or sudden traffic congestion [9]. In such contexts, UAVs exhibit notable flexibility and responsiveness, especially in regions lacking adequate terrestrial infrastructure [10]. Nevertheless, despite their advantages, UAVs are constrained by limited onboard energy, which significantly hampers their ability to sustain continuous operations during extended missions. When deployed in remote or hard-to-reach areas for tasks such as sensor data acquisition and distributed mission collection, efficient task execution under strict energy constraints becomes critical [11]. Therefore, optimizing UAV deployment and trajectory planning is essential to reducing flight energy consumption and ensuring successful mission completion.

Given the rapid development of mobile terminals, a single UAV may suffer from inefficiency or task failure due to limited onboard energy during mission execution. To address this issue, this article proposes an adaptive clustering approach to enable the rational deployment of multiple UAVs, thereby alleviating the workload on individual UAVs. Tasks are assigned based on the performance characteristics of each UAV, significantly improving overall operational efficiency. Furthermore, a deep reinforcement learning (DRL) algorithm that integrates a pointer network with attention mechanisms is developed to optimize UAV trajectories effectively, ultimately generating collaborative multi-UAV trajectory strategies that cover all designated task regions.

II. RELATED WORKS

Current UAV trajectory design strategies are commonly classified into two major categories: conventional approaches (such as exact, heuristic, and metaheuristic methods) and machine learning-based techniques. Although precise algorithms offer optimal results via formal mathematical modeling or exhaustive enumeration, they suffer from high computational costs as the problem size increases, which limits their practical applicability [12]. For instance, geometry-based methods construct constraint models based on known environments and utilize algorithms such as A^* , Dijkstra, and Voronoi diagrams to find feasible paths, particularly in static settings. However, in real-world scenarios where environmental information is often unavailable or delayed, the practicality and accuracy of such methods are significantly constrained.

Heuristic and metaheuristic algorithms for UAV path planning can be further classified into three main categories within stochastic metaheuristic methods: evolutionary strategies, physical-model-driven techniques, and bio-inspired

swarm approaches. Representative evolutionary strategies include genetic algorithms and differential evolution [13], which evolve populations iteratively via genetic variation mechanisms such as crossover and mutation. Physical-model-driven methods, such as artificial potential fields and simulated annealing [14], draw inspiration from natural physical processes. Swarm intelligence methods, such as ant colony systems and particle swarm optimization [15], are extensively employed in UAV trajectory design by emulating the cooperative dynamics observed in biological swarms. However, these methods do not guarantee solutions that are close to the global optimum. However, with the advancement of RL technologies, increasing attention has been directed toward their application in addressing UAV trajectory planning problems. For example, Abedin et al. [16] proposed a DRL model with experience replay, which considers contextual constraints such as energy consumption and information freshness to address the trajectory policy learning problem. To ensure efficient and fair communication services, Ding et al. [17] proposed a DRL-based joint optimization algorithm for 3-D UAV trajectory planning and frequency band allocation, built upon the deep deterministic policy gradient (DDPG) framework, which improves system throughput while considering energy consumption constraints. Yuan et al. [18] proposed an actor-critic-based deep stochastic online scheduling algorithm to optimize data transmission scheduling and hovering time in UAV-assisted communication networks, achieving near-optimal energy savings while significantly reducing computational complexity.

For combinatorial optimization challenges like the traveling salesman problem and vehicle routing problem, these tasks are often reframed in machine learning as sequential decision-making processes. Therefore, the design of machine learning algorithms for such problems must demonstrate strong capabilities in learning strategies over sequential data. However, algorithms such as convolutional neural networks (CNNs), deep Q-networks (DQNs), and DDPG often struggle with sequence tasks where current decisions are heavily influenced by prior contextual information. These methods struggle to retain long-term memory of past elements, which is particularly crucial for path optimization tasks where information about previously visited nodes is essential. Long short-term memory (LSTM) networks are commonly employed for sequence modeling due to their ability to store long-range historical information via gated memory units. Similarly, recurrent neural networks (RNNs) process sequences of varying lengths by sharing network components and weights across time steps, making them well-suited for modeling variable-sized input data [19]. Attention mechanisms offer another effective approach for processing variable-length sequences by allowing models to dynamically focus on relevant parts of the input through shared parameters. Consequently, RNN-based and attention-based Seq2Seq models—typically consisting of an encoder-decoder architecture—have emerged as attractive solutions for variable-length sequence problems [20]. These techniques have been successfully applied across various domains. For instance, in network function virtualization (NFV) infrastructure, attention-based Seq2Seq models

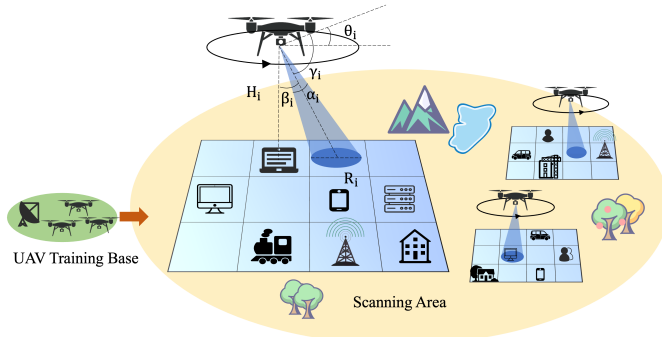


Fig. 1. Multi-UAV regional task search and trajectory optimization.

combined with reinforcement learning have been used to optimize resource placement strategies for required network services, achieving substantial reductions in overall energy consumption [21].

Compared to the aforementioned studies, this article proposes an integrated framework that significantly enhances UAV task efficiency in complex search environments. By incorporating UAV capability assessment, adaptive clustering-based deployment, and DRL for trajectory optimization, the framework enables efficient and coordinated task execution among heterogeneous UAVs.

III. SYSTEM MODEL AND PROBLEM FORMULATION

A. UAV Model

This article investigates a trajectory planning problem for cooperative multi-UAV task search. As illustrated in Fig. 1, a fleet of m UAVs is considered, represented by the set $U = \{U_1, U_2, \dots, U_m\}$, which collaboratively search across n regions $A = \{A_1, A_2, \dots, A_n\}$. Given the heterogeneity of UAVs, each UAV is characterized by distinct capabilities and attributes, such as energy consumption, flight speed, coverage range, and endurance. All UAVs are initially assumed to be deployed from a shared training base. The model focuses solely on UAV mobility, and scanning capabilities are considered, while structural factors like geometry, dimensions, and internal structure are omitted. Based on this abstraction, the study focuses on optimizing UAV trajectories to improve task execution efficiency.

The UAV is represented as a vector $U_i = \{v_i, v_i^s, H_i, \theta_i, \alpha_i, \beta_i, \gamma_i, q_i\}$, where v_i denotes the UAV's average flight speed, v_i^s denotes its scanning speed, H_i signifies the flight altitude, and θ_i is the angle between the UAV's flight direction and the horizontal plane (positive for upward flight and negative for downward flight). Fig. 1 illustrates the detection range of the sensors on UAV U_i . α_i denotes the sensor's scanning angle, β_i stands for the angle between the sensor and the vertical ground, γ_i is the sensor's installation angle, and q_i signifies the average power consumption of the UAV during flight. In this study, it is assumed that the UAV's flight altitude depends solely on its intrinsic properties and remains constant during task execution. Similarly, the sensor's installation and scanning

angles are considered fixed. Therefore, the scanning width R_i of UAV U_i can be calculated as follows:

$$R_i = 2 \times \tan \frac{\alpha_i}{2} \times \frac{H_i}{\cos \beta_i} = 2 \times \tan \frac{\alpha_i}{2} \times \frac{H_i}{\sin(\gamma_i - \theta_i)}. \quad (1)$$

A larger scanning width, R_i , signifies that the UAV has a faster scanning speed and, consequently, better performance.

B. UAV Task Collection Decision

The UAV task collection strategy refers to the UAV's scanning trajectory over region A . To record the UAVs' flight paths during the task collection process, a 3-D array $X = \{x_{i,j,k}\}, i \in [1, m], j, k \in [0, n]$ is employed. The variable $x_{i,j,k}$ is defined to represent whether UAV U_i follows a flight path from area A_j to area A_k . If this travel route exists, then $x_{i,j,k} = 1$; if not, $x_{i,j,k} = 0$. Consequently, we can derive:

$$x_{i,j,k} = \begin{cases} 1, & \text{if UAV } U_i \text{ travels from } A_j \text{ to } A_k \\ 0, & \text{else.} \end{cases} \quad (2)$$

In the context of performing search tasks, this article considers that the number of UAVs executing tasks should be less than the total number of UAVs available. All UAVs, denoted as U , take off from the same training base represented by A_0 . The value of $x_{i,0,j}$ indicates whether there exists a flight trajectory from the UAV training base A_0 to region A_j . This condition is formulated as follows:

$$\sum_{j=1}^n x_{i,0,j} \leq m \quad \forall i \in [1, m]. \quad (3)$$

Considering that multiple UAVs may search simultaneously, there is a potential risk that more than one UAV could attempt to scan the same region. To prevent such conflicts and avoid collisions, each region must be assigned to exactly one UAV. Accordingly, for any region A , only one UAV is allowed to receive the task, which leads to the following constraint:

$$\sum_{i=1}^m \sum_{k=1}^n x_{i,j,k} = 1, \sum_{i=1}^m \sum_{k=0}^n x_{i,k,j} = 1 \quad \forall j \in [1, n]. \quad (4)$$

During the UAV task collection process, UAVs are required to systematically scan and collect data from the task regions. The array $Z = \{z_{i,j}\}, i \in [1, m], j \in [0, n]$, is used to record the visitation order of each region, where each element $z_{i,j}$ denotes the sequence index at which UAV U_i visits region A_j . Since all UAVs take off from the same training base A_0 , the departure point A_0 is assigned an index of 1, i.e., $z_{i,0} = 1, i \in [1, m]$. After UAV U_i completes scanning region A_j and proceeds to region A_k for continued data collection, the visitation order of A_k relative to A_j can be expressed as $z_{i,k} = z_{i,j} + 1$, from which it follows that $x_{i,k,j} = 1$. Therefore, the visitation order can be described as:

$$\begin{cases} z_{i,0} = 1 \\ z_{i,k} = z_{i,j} + 1, & \text{if } x_{i,k,j} = 1. \end{cases} \quad (5)$$

The number of scanning regions assigned to each UAV is constrained by the total number of regions. Therefore, the visitation index for any region must be less than or equal to $n + 1$. This constraint can be expressed as

$$z_{i,j} \leq n + 1 \quad \forall i \in [1, m] \quad \forall j \in [0, n]. \quad (6)$$

C. Cost Model

The cost model includes both the UAV's scanning duration cost and the flight cost. For each region A_j , denoted by S_j , representing the area to be scanned, and assuming UAV U_i conducts the scanning, the time cost within A_j , denoted as $T_i^{S_j}$, and the flight cost of the UAV, denoted as $p_i^{S_j}$, can be calculated as follows:

$$T_i^{S_j} = \frac{S_j}{R_i \times v_i^s} \quad (7)$$

$$p_i^{S_j} = q_i \times T_i^{S_j}. \quad (8)$$

Considering that the n regions are mutually separated, a 2-D array $D = \{D_{j,k}\}, j, k \in [1, n]$ is adopted to record the distance between any two regions. The time cost and flight cost incurred by UAV U_i when traveling from region A_j to region A_k are calculated as follows:

$$T_{j,k}^{U_i} = \frac{D_{j,k} \times x_{i,j,k}}{v_i} \quad (9)$$

$$p_{j,k}^{U_i} = q_i \times T_{j,k}^{U_i}. \quad (10)$$

It is important to note that if there is no flight trajectory between region A_j and A_k for UAV U_i , then there are no time costs or flight costs associated with this scenario.

In this work, it is essential to ensure that each UAV can successfully complete its assigned scanning task. Therefore, the total energy consumption for both flight and scanning must not exceed the UAV's available energy. By combining (8) and (10), this constraint can be formulated as

$$p_i^{S_j} + p_{j,k}^{U_i} \leq p_{U_i}. \quad (11)$$

D. Problem Formulation

The objective of this work is to determine the optimal trajectories for multiple UAVs in order to minimize the overall task cost. The total cost is defined as a weighted sum of the total time required and the total energy consumed by the UAVs to complete their assigned tasks. In particular, $T(U_i)$ enotes the time consumed by UAV U_i to scan its assigned regions, while $P(U_i)$ represents the corresponding energy consumption, i.e.,

$$T(U_i) = \sum_{j=0}^n \sum_{k=1}^n T_{j,k}^{U_i} + T_i^{S_j} \quad (12)$$

$$P(U_i) = \sum_{j=0}^n \sum_{k=1}^n p_{j,k}^{U_i} + p_i^{S_j}. \quad (13)$$

In multi-UAV systems, the overall task execution efficiency is determined by the UAV with the latest completion time. Consequently, the optimization objective is defined as follows:

$$\begin{aligned} \min & \left(\sigma \sum_{i=1}^m P(U_i) + \max T(U_i) \right), \quad i \in [1, m] \\ \text{s.t.} & \text{ (3), (4), (5), (6), (11)} \end{aligned} \quad (14)$$

where σ is a weighting factor of energy consumption relative to task completion time. It serves to normalize and balance the two terms so that neither dominates the objective function. Its value can be set according to the priority of the specific task.

For example, a higher σ is used in scenarios emphasizing low-energy consumption, whereas a smaller σ is preferred in time-critical missions.

IV. ADAPTIVE TRAJECTORY OPTIMIZATION METHOD

In this section, we propose a UAV capability-aware adaptive route optimization (CAR) algorithm. Given the heterogeneity in UAV capabilities and attributes, a performance evaluation is first conducted for each UAV. UAVs with higher speed and stronger scanning capabilities are assigned to regions with heavier workloads, while those with lower performance are tasked with lighter workloads. Subsequently, an adaptive clustering algorithm guided by spatial density and interregional distance is devised to autonomously group the task areas and assign scanning regions to each UAV, thereby effectively addressing the challenge of automated path planning in multi-UAV systems. This clustering process sequentially groups areas based on regional density and allocates one cluster to each UAV, ensuring a balanced workload distribution. Finally, a pointer network is integrated by leveraging a reinforcement-driven decision-making framework for inferring the most efficient sequence for regional traversal, thereby minimizing the overall cost of multi-UAV scanning operations.

A. Assessment of UAV Capabilities

In this study, the assessment of UAV capabilities plays an indispensable role in the trajectory optimization scheme. It facilitates the rational allocation of different-function UAVs and prioritizes them based on their capabilities for the subsequent assignment of scanning areas. When UAVs are executing collection tasks, considering the varying attributes of UAVs, such as scanning capacity, flight speed, and total energy consumption, it becomes essential to evaluate their capabilities to facilitate appropriate task assignment. This work proposes a metric aimed at evaluating UAV performance, and high-capability UAVs are assigned areas with intensive search tasks. The capability assessment function for UAV U_i is represented as $F(U_i)$. It is utilized exclusively when employing UAV U_i for task collection to achieve predefined objectives and determine the total cost. The capability division is determined based on the total cost and is formulated as

$$F(U_i) = \delta \sum_{j=1}^{n^*} (T_i^{S_j} + p_i^{S_j}) + (1 - \delta) \sum_{j=0}^{n^*} \sum_{k=1}^{n^*} (T_{j,k}^{U_i} + p_{j,k}^{U_i}) \quad (15)$$

where $\delta \in [0, 1]$ is a predefined weighting factor used to balance scanning and flight performance, and n^* is a number smaller than n that controls the size of the task region subset used in the capability evaluation. In each experimental group, the selected n^* and its corresponding region subset are kept fixed and applied uniformly to all UAVs for consistent capability assessment. As shown in (15), each UAV's capability score is determined based on the expected total cost incurred during task execution, which includes both scanning time and energy cost, as well as flight time and energy cost. UAVs with a lower total cost are considered more capable. In this study, we set $\delta = 0.5$ to maintain a neutral balance between scanning

and flight performance. Based on the clustering results, UAVs are deployed accordingly, and tasks are assigned based on their evaluated capabilities to improve the overall efficiency of trajectory optimization.

B. Region Partitioning Based on Density and Distance Adaptive Clustering

The clustering-based region partitioning problem is a multiconstrained combinatorial optimization task, whose solution space and computational complexity increase exponentially with the total number of regions. In this study, a novel clustering method is proposed that does not require predefining the number of clusters. Instead, it adaptively partitions the regions based on interregional distances and local density information. In particular, the local density of each region is first calculated. Here, “density” does not simply refer to the task intensity within the region, but is defined as the sum of UAV flight time costs to neighboring regions located within a predefined cutoff distance. In addition, the minimum distance from each region to any other region with a higher local density is computed to assist in identifying cluster centers. A region is designated as a cluster center if it has a significantly higher local density than its surrounding regions and is also relatively distant from other centers. Subsequently, the remaining regions are assigned to the cluster associated with the nearest center. To further describe the density calculation for a region, let’s consider region A_j as an example. The density of region A_j , denoted as φ_j , can be computed as follows [22]:

$$\varphi_j = \sum_{k=1}^n \chi(D_{j,k}, d_c) \left(T_i^{S_k} + T_i^{S_j} + \lambda T_{j,k}^{U_i} \right). \quad (16)$$

This approach facilitates automatic region clustering based on both density and distance metrics, without the need for specifying the number of clusters in advance. d_c is the cutoff distance of a region, serving as an algorithmic neighborhood-scale parameter to define the spatial range within which regions are regarded as closely related for local density computation and cluster formation. $T_i^{S_j}$, $T_i^{S_k}$, and $T_{j,k}^{U_i}$ stand for the average scanning time for A_j , the average scanning time for A_k , and the average flight time from A_j to A_k , respectively. Furthermore, λ is a weight parameter signifying the relative importance of flight time in comparison to scanning time. The function $\chi(D_{j,k}, d_c)$ is a Boolean function designed to determine whether the distance between region A_k and region A_j is smaller than the truncation distance d_c . It can be expressed as follows:

$$\chi(D_{j,k}, d_c) = \begin{cases} 1, & \text{if } D_{j,k} \leq d_c \\ 0, & \text{else.} \end{cases} \quad (17)$$

To avoid boundary discontinuities caused by a hard threshold, d_c is determined by considering the distribution of interregional distances and UAV flight characteristics, ensuring that small perturbations do not materially affect the clustering results. In terms of computational complexity, since pairwise distances and densities must be computed for n regions, this process has a time complexity of $O(n^2)$. Nevertheless, this module is executed once during the system initialization phase

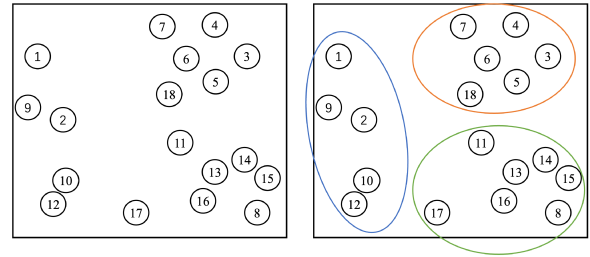


Fig. 2. Regional distribution and division of adaptive clustering.

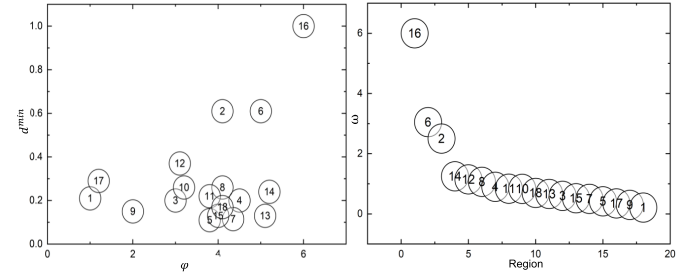


Fig. 3. Regional decision map.

as an offline preprocessing step, and its computational cost is manageable in practical deployments.

Taking region A_j as an example, the shortest distance between this region and any other region exhibiting a greater density is denoted as $d_j^{\min}$. In particular, if $\varphi_k > \varphi_j$, then $d_j^{\min} = \min(D_{j,k})$. In this study, the selection of cluster centers is based on both local density and the corresponding minimum distance. As illustrated in the left panel of Fig. 2, a 2-D map is generated with 18 randomly distributed regions, each ranging from 50 to 100 m², for clustering analysis.

After obtaining the local density φ_j of each region and its minimum distance $d_j^{\min}$ to any region with higher density, we define the importance measure $\omega_j = \varphi_j d_j^{\min}$ to balance density magnitude and spatial separability when identifying potential cluster centers. Regions with larger ω_j values are more likely to serve as cluster centers. As illustrated in Fig. 3, the left side of Fig. 3 presents the decision graph constructed from φ_j and $d_j^{\min}$, while the right side of Fig. 3 shows the regions sorted in descending order of ω_j . In typical scenarios, task regions exhibit natural spatial aggregation, resulting in a pronounced jump in the sorted ω_j sequence that clearly separates cluster centers from noncenter regions. To avoid the uncertainty of manual identification when the distribution of regions is uniform or contains noise, we further introduce an automated thresholding mechanism. In particular, by applying the Otsu adaptive thresholding method to the adjacent-difference sequence of the sorted ω_j values, the jump point can be automatically detected, enabling robust and data-driven selection of cluster center candidates.

It is worth noting that the design of ω_j balances local density and spatial separation. A higher φ_j indicates that a region and its neighbors have stronger task intensity and correlations, while a larger $d_j^{\min}$ ensures that the candidate center is well separated from denser regions, preventing overly concentrated cluster centers. Compared with single-factor metrics that rely only on density or distance, ω_j achieves

a better tradeoff between representativeness and distribution, which enhances clustering stability and improves overall task allocation efficiency. This idea is consistent with classical density-peak clustering principles and shows clear advantages in our experiments.

Once the cluster centers are identified, regions with higher ω values among the cluster centers are allocated higher capability drones for collecting tasks, as per the calculation of drone capability levels in (15). The remaining cluster center regions are assigned drones based on their capability assessment and ω values. Through this clustering scheme, drones are systematically allocated to the required regions based on their capabilities, ensuring a fair correspondence with the cluster centers. The right side of Fig. 2 illustrates the clustering results for the regions on the left side.

C. Pointer Network Architecture for UAV Trajectory Optimization

Through adaptive clustering, a set of region clusters is obtained and denoted as $G = \{G_1, G_2, \dots, G_m\}$. Where each UAV is responsible for sequentially visiting the areas within its assigned cluster to collect data. Therefore, the trajectory planning task can be transformed into a sequential decision-making problem over the assigned regions. In this study, we focus on a single cluster for detailed analysis. The corresponding trajectory planning strategy is modeled using a pointer network, where the encoder processes the input regions within the cluster, and the decoder generates the visiting sequence as output. Given the UAV's initial position A_0 and n task regions, the input to the model is defined as $C = \{A_0, A_1, A_2, \dots, A_n\}$, and the output is a permutation representing the visiting order, denoted by $Y = \{\pi_0, \pi_1, \pi_2, \pi_3, \dots, \pi_n\}$. An attention mechanism is employed as a pointer to select elements from the input sequence as the output. Since the UAV departs from A_0 , it must appear as the first element in the output sequence Y . For notational consistency, we denote π_0 as the index corresponding to A_0 in Y . The overall probability of sequence Y , which represents the likelihood that the UAV follows the corresponding trajectory, can be factorized using the chain rule as follows:

$$P_\theta(Y|C) = \prod_{t=0}^K P(\pi_t | \pi_0, \dots, \pi_{t-1}, C) \quad (18)$$

where t represents the current time index. The policy $P_\theta(Y|C)$, governed by parameters θ , represents the probabilistic strategy responsible for determining the visiting sequence. $P(\pi_t|\cdot)$ models the probability of selecting any region at time step t , conditioned on the set of regions visited during the preceding $t - 1$ time steps.

As illustrated in Fig. 4, a pointer network is employed to model the conditional probability $P(\pi_t|\cdot)$. Traditional Seq2Seq neural networks consist of an encoder and a decoder. The encoder processes the input sequence into a fixed-dimensional representation, and the decoder generates the corresponding output sequence. In this context, the pointer network is used to produce position indices, allowing it to generate the access order of the input regions [23].

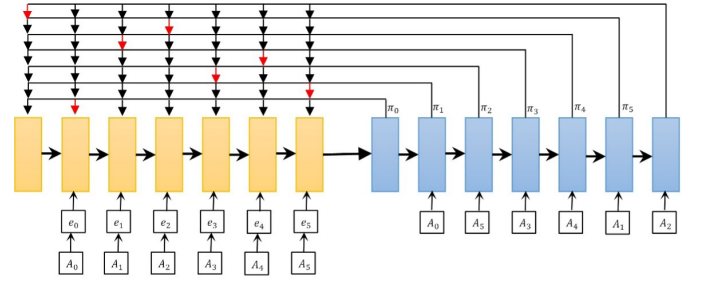


Fig. 4. Pointer network model.

1) *Encoder*: Traditional RNNs struggle with modeling long-term dependencies; therefore, the pointer network adopts LSTM units to construct a recurrent encoder capable of effectively capturing long-range temporal relationships. The encoder input consists of the coordinates of all region nodes, which inherently do not possess a predefined ordering. Consequently, the model treats them as an unordered set and maps each region to an independent feature vector through an embedding matrix. The spatial coordinates of each node are directly preserved within its embedding, and during the subsequent attention computation, these embeddings interact with the decoder's hidden state to model interregion correlations. This allows the network to implicitly learn geometric relationships and spatial proximity among regions, enabling pointer networks to rely on such implicit spatial modeling without requiring explicit positional encoding. Compared with the more lightweight gated recurrent unit (GRU), the additional gating mechanism of LSTM offers a stronger ability to preserve long-term dependencies. This design effectively reduces the computational complexity of the embedding layer without sacrificing overall efficiency and fully meets the requirements of modeling variable-length sequences. The transformation of input C into its embedded representation $\tilde{C}$ is formulated as follows:

$$\tilde{C} = E_b C \quad (19)$$

where E_b represents the embedding matrix, and $\tilde{C} = \{e_i\}_{i=0}^n$. As an illustration, consider a case with one initial location and five task areas, forming the input set $C = \{A_0, A_1, A_2, A_3, A_4, A_5\}$. After embedding, C is transformed into $\tilde{C} = \{e_0, e_1, e_2, e_3, e_4, e_5\}$.

2) *Decoder*: In the pointer network, the decoder RNN is also implemented using LSTM units. The output of the encoder $\tilde{C} = \{e_0, e_1, e_2, e_3, e_4, e_5\}$ is then fed into the decoder. The total number of decoding iterations corresponds to the size of $\tilde{C}$. During each decoding iteration t , the LSTM generates a hidden representation h_t that captures contextual information from prior steps. Based on the attention mechanism, the decoder employs h_t along with the embedded inputs $\tilde{C}$ to derive the conditional probability $P(\pi_t | \pi_0, \dots, \pi_{t-1}, C)$, which guides the selection at the current step. The attention mechanism is responsible for computing this probability, with further elaboration provided later in this section.

3) *Attention Mechanism*: This mechanism mainly serves to strengthen the interaction between the encoder and decoder by dynamically weighting input elements based on their relevance. In the context of UAV trajectory planning, it enables

the model to capture the relevance between each candidate region at decoding step t and the previously visited regions up to step $t - 1$. At each step, the region with the highest relevance, reflected by the highest probability, is selected as the next target in the UAV's path. Consequently, u_t^i is derived using the following formulation:

$$u_t^i = \begin{cases} v^T \tanh(W_1 e_i + W_2 h_t), & \text{if it's valid} \\ -\infty, & \text{else} \end{cases} \quad (20)$$

where e_i denotes the encoder's output at step i , representing its internal state, while h_t represents the decoder's hidden state at time step t . The parameters v^T , W_1 , and W_2 are collectively denoted as θ and serve as learnable weights in the pointer network model. u_t^i denotes the relevance of step t to region e_i . The conditional probability is calculated by applying the softmax operation to the relevance values of all regions yet to be visited, as shown below

$$P(\pi_t | \pi_0, \dots, \pi_{t-1}, C) = \text{softmax}(u_t^i). \quad (21)$$

This study employs a pointer network framework to tackle the UAV trajectory design challenge. Typically, UAV trajectory optimization involves finding an effective policy for a constrained decision-making problem, aiming to minimize task completion time or energy consumption, thereby reducing the overall operational cost. In this context, ground clusters, performance targets, and imposed constraints are regarded as part of the environment. By learning an optimal policy that maximizes the cumulative reward, the UAV can determine an efficient trajectory. An actor-critic approach is adopted for model training, where the UAV serves as an agent engaging with the environment in a stepwise manner. Consequently, the UAV trajectory optimization problem is formulated within the framework of a Markov decision process (MDP).

The MDP is a mathematical framework used to model sequential decision-making problems, and it is widely applied in reinforcement learning and optimal control. An MDP typically consists of the following components: a state space S , which characterizes the environment at any given time; an action space $\mathcal{A}$, which denotes the set of possible actions that an agent can take in each state; a state transition probability $\mathcal{P}$, which defines the probability of transitioning from state s to state s' after taking action a ; and a reward function R , which provides the immediate feedback received after executing action a in state s . The MDP formulation of the UAV trajectory optimization problem is defined as follows:

4) *State Space S* : The state at time step t is composed of the coordinates of all regions, the location of the drones, time costs, and energy costs of the drones.

5) *Action Space $\mathcal{A}$* : At each time step t , the UAV's action is to select the next region to visit. Therefore, the outputs of the attention mechanism and region selection are defined as actions at each step.

6) *Transition Probabilities $\mathcal{P}$* : The transition probability is determined by the attention mechanism, which selects the most relevant region with the highest probability as the next region to visit. If an action would violate spatial constraints—such as selecting an already assigned region or attempting to reach a region with insufficient remaining energy—the action is

treated as infeasible and removed from the candidate set. This mechanism allows spatial constraints and collision-avoidance logic to be explicitly encoded at the policy level.

7) *Reward Function R* : The reward function is defined as the opposite of the total cost incurred. The reward value under the given policy is calculated as $R = -(\sigma \sum_{i=1}^m P(U_i) + \max T(U_i))$. σ is a predefined weight coefficient.

Although the pointer network generates a local trajectory for a single region cluster C at each training iteration, the optimization objective of the multi-UAV cooperative task is inherently global, with the reward determined jointly by the total energy consumption of all UAVs and the overall mission completion time. Therefore, the training process employs a unified set of shared actor-critic network parameters across all clusters and updates the policy using the global reward R . Each cluster thus serves only as a subproblem instance for training, rather than as an independent optimization unit.

The actor-critic algorithm integrates the advantages of both policy gradient methods (Actor) and value-based methods (Critic), offering improved convergence and enhanced training stability. In our framework, a pointer network is employed as the actor network, aiming to reduce gradient variance during training and thereby enhance learning efficiency. A RNN, composed of LSTM units, is adopted as the critic network. Given the known input, it predicts the expected reward based on the final state generated by the actor network. For a given problem instance C , the training objective is to maximize the expected return, defined as

$$J(\theta|C) = \mathbb{E}_{Y \sim p_{\theta}(Y|C)} [R]. \quad (22)$$

In order to optimize θ , a policy gradient method is adopted, which computes the derivative of the expected reward concerning the parameter θ . This gradient is typically estimated using stochastic gradient descent (SGD) during training. Using the strategy gradient algorithm to represent (22)

$$\nabla_{\theta} J(\theta) \approx \frac{1}{B} \sum_{i=1}^B (R_i - V_{\psi}(C_i)) \nabla_{\theta} \log p_{\theta}(Y_i|C_i). \quad (23)$$

where B denotes the total number of problem instances contained in each training batch. The critic network is trained by employing SGD to minimize the mean squared error between the predicted value $V_{\psi}(C_i)$ and the actual value as the optimization objective, which is expressed as

$$\mathcal{L}(\psi) = \frac{1}{B} \sum_{i=1}^B (V_{\psi}(C_i) - R_i)^2. \quad (24)$$

Algorithm 1 demonstrates the training procedure of the actor and critic networks, where the actor-critic parameters are progressively optimized via the Adam method. The proposed model, comprising the pointer network, actor network, and critic network, exhibits strong generalization capabilities. During the training phase, the model can be trained on small-scale problem instances and subsequently generalized to solve larger instances.

TABLE I
UAV PARAMETERS

UVA number	1	2	3	4	5	6	7	8	9	10
Flight Speed	6.0	5.0	5.5	7.0	8.5	4.0	7.5	8.2	6.5	5.5
Flight Altitude	22	21	18	21	20	17	19	25	18	20
Flight Energy Consumption	45	40	43	50	55	40	52	50	48	45
θ_i	8.0	7.5	9.0	8.5	6.5	7.5	5.5	6.5	6.0	7.0
α_i	11.0	10.0	10.5	11.5	10	9.5	12	13.5	12	12.5
γ_i	81	69	79	80	87	82	83	76	86	82

Algorithm 1 CAR Algorithm

Input: A set of UAVs $U = \{U_1, U_2, \dots, U_m\}$, a set of regions $A = \{A_1, A_2, \dots, A_n\}$, batch size B , number of training steps S

Output: Flight trajectories for UAVs

UAV Capability Assessment:

- 1: Assume a task collection scenario and define a region set A^1 (independent of A);
- 2: Calculate the UAV's capability level using Equations (15);
- 3: Sort the UAV capability levels in ascending order. **Region Partitioning:**
- 4: Compute φ_j and d_j^{min} for each region A using Equations (16) and (17);
- 5: Calculate the ω_j value for each region;
- 6: Select cluster centers based on the ω_j values using the Otsu adaptive thresholding method;
- 7: Match the UAV levels obtained from the UAV capability assessment algorithm with the ω_j values of cluster centers.

Training the Actor-Critic Algorithm for the Pointer Network:

- 8: Initialize the θ and ψ parameters in the actor and critic networks with random weights;
- 9: **for** $s = 1$ to S **do**
- 10: Transform the region set C into $\tilde{C}$ using Equation (19);
- 11: Sample from $\tilde{C}$ and compute Y and R using the pointer network;
- 12: Calculate $V_\psi(C_i)$ using the critic network;
- 13: $d\theta \leftarrow \frac{1}{B} \sum_{i=1}^B (R_i - V_\psi(C_i)) \nabla_{\theta} \log p_{\theta}(Y_i | C_i)$;
- 14: $\mathcal{L}(\psi) \leftarrow \frac{1}{B} \sum_{i=1}^B (V_\psi(C_i) - R_i)^2$;
- 15: $\theta \leftarrow \text{Adam}(\theta, d\theta)$;
- 16: $\psi \leftarrow \text{Adam}(\psi, \nabla_{\psi} \mathcal{L}(\psi))$;
- 17: **end for**
- 18: **return** the trained parameters θ .

$v_i^s = \mu * v_i$, where μ is a system resistance factor significantly affecting the UAV's scanning speed. The experimental UAV configurations are summarized in Table I, where the weight parameter λ is assigned a value of 1. The n regions are subjected to adaptive clustering, resulting in m clusters. The problem is then transformed into the allocation of UAVs from the initial position A_0 to m clusters and trajectory optimization. All generated regions constitute the final training dataset. The model is trained for 10 000 steps in 20 different regions and 50 different regions, with a batch size of 64 for each training step. The actor and critic networks each contain 128 neurons, with an initial learning rate set to 0.001 and a discount factor of 0.96.

To comprehensively evaluate the performance of the proposed CAR algorithm, we conducted comparative experiments with several representative methods. All methods were reimplemented under a unified simulation environment to ensure fair comparison, using consistent energy consumption models, distance metrics, and UAV configuration parameters.

1) *Greedy Algorithm*: The greedy search strategy involves selecting the region with the highest probability for task processing throughout the entire reasoning process, denoted as "Greedy" in the simulation experiments.

2) *Ant Colony Optimization*: In the ACO algorithm, ants select paths with the highest concentration of pheromones. Specific parameter settings include 30 ants, 200 iterations, an evaporation rate of 0.1 for pheromones, and relative importance values of 1 for pheromones and 5 for visibility. It is labeled as "ACO" in the simulation experiments.

3) *K-Means-CA-TSP*: The K -means-CA-TSP algorithm initially partitions tasks within regions using K -means and then proposes an iterative UAV trajectory optimization algorithm inspired by the capacitated arc-TSP (CA-TSP) [24]. It is denoted as "KCA-TSP" in the simulation experiments.

4) *STCA-No*: The STCA-NO algorithm introduces a density-based clustering method for region partitioning and subsequently optimizes the order using a genetic algorithm [25]. It is denoted as "STCA-NO" in the simulations.

V. EXPERIMENT AND DISCUSSION

A. Parameter Settings

This article considers a grid with a size of 2×2 km for the ground area. The UAV's initial position A_0 set at (0 and 0 m). The problem addressed in this article involves collecting tasks from n regions, and the goal is to minimize the collection cost through drone deployment optimization and drone trajectory optimization. To start, n regions are generated, and their utilization rates τ_i with respect to the total scanning area A_i of each region are calculated. This allows us to obtain $S_i = \tau_i * A_i$, where A_i represents the total area of region i . The scanning speed of the UAVs in region A_i can be calculated as

B. Experimental Results Analysis

Fig. 5 illustrates the variation of the policy gradient $\nabla_{\theta} J(\theta)$ with respect to the number of iterations, depicting the convergence behavior of the actor network. It can be observed that the gradient gradually approaches zero after approximately 2000 iterations, indicating that the algorithm exhibits favorable stability and convergence performance.

The impact of the number of regions on the total cost is evaluated. Fig. 6 shows that the total cost increases as the

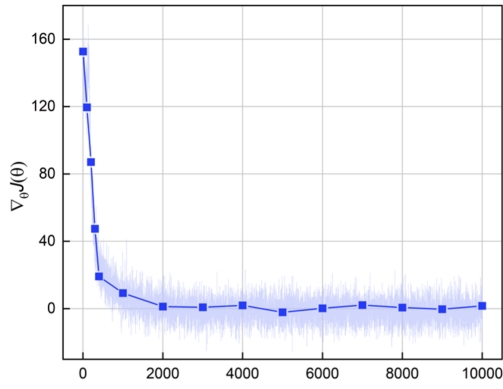


Fig. 5. Training curve.

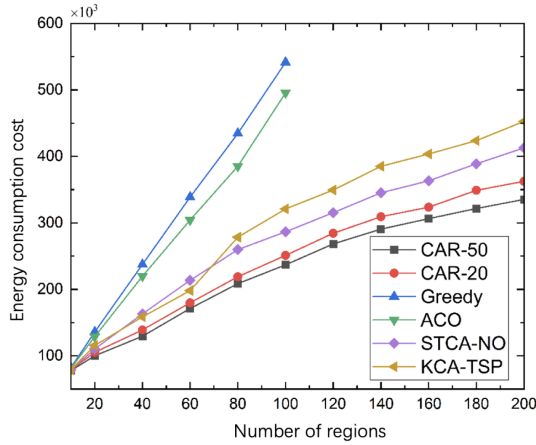


Fig. 6. Impact of the number of regions on the total cost.

number of regions grows, mainly because more UAVs must be deployed and their flight distances become longer, leading to higher energy consumption. When the number of regions remains fixed, our proposed method achieves the lowest total cost in most cases. For example, when the number of regions reaches 100, the energy cost of CAR-50 is 6% lower than that of CAR-20, and the total cost of our algorithm is 53.6 %, 49.3%, 21%, and 12% lower than those of the greedy search algorithm, ant colony optimization, *K*-means-CA-TSP, and STCA-NO, respectively. It can be observed that the greedy search and ant colony optimization algorithms assume a single UAV performs all tasks; when the number of regions is large, the UAV's processing cost becomes excessive, causing tasks to fail due to its own energy limitations. It should also be noted that the main computational load of our method comes from pairwise calculations of interregional distances and local densities, which scale approximately quadratically with the number of regions, while the subsequent cluster-center selection and UAV assignment steps grow roughly linearly. Consequently, the overall computational complexity remains acceptably scalable as the number of regions increases.

Furthermore, an evaluation was conducted to examine how varying the number of regions affects the UAV task completion time. As illustrated in Fig. 7, the completion time of all five methods increases as the number of regions continues to grow. Notably, it's observed that the greedy algorithm and

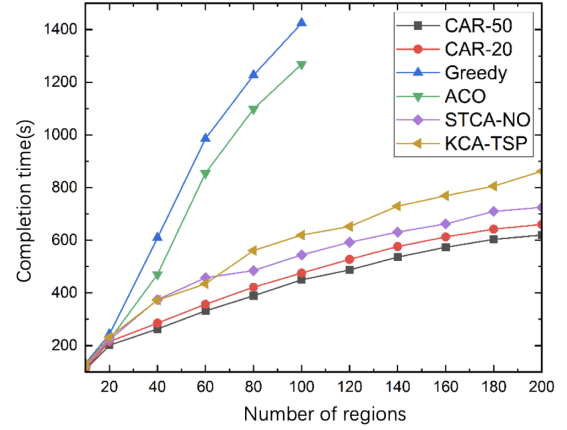
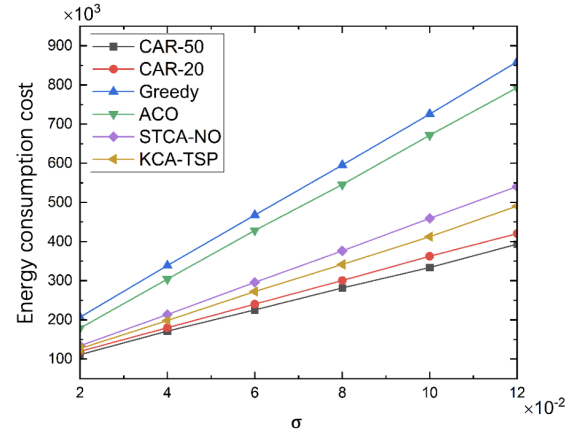


Fig. 7. Impact of the number of regions on task completion time.

Fig. 8. Impact of σ changes on the total cost.

ACO algorithm, due to their single-drone implementation, experience significantly longer completion times. In contrast, methods like KCA-TSP, STCA-NO, and CAR, which deploy multiple drones, outperform single-drone approaches, yielding faster task completion times. With an increasing number of regions, the CAR-50 method showcases a 5.7% improvement in performance compared to CAR-20. Furthermore, it achieves a reduction in completion time of 21% and 8% compared to the other two methods.

This article also investigates the impact of the weight parameter σ on the system's cost, which represents the tradeoff between power usage and task duration. As depicted in Fig. 8, with an increase in the weight parameter σ , the CAR-50 and CAR-20 algorithms exhibit the least steep slopes compared to other methods. These slopes symbolize the UAV energy consumption rate. Therefore, appropriate UAV deployment and path planning can effectively lower total UAV power consumption.

VI. CONCLUSION

In this article, we design a framework for UAV deployment and trajectory optimization. First, a UAV capability

evaluation method is proposed to reasonably allocate service regions among UAVs with different performance levels. Then, an adaptive clustering approach is employed to address the regional deployment problem. Finally, a pointer network combined with DRL is used to optimize UAV trajectories. A key advantage of the proposed DRL algorithm is its ability to generalize beyond the regions used for training. The model can be trained quickly on small-scale clusters and applied to large-scale problems. Simulation results confirm its consistent superiority over baseline methods, achieving the best performance across multiple evaluation metrics. In future work, we plan to extend the proposed framework to handle more dynamic scenarios, such as online region updates and variable UAV starting positions.

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